Fitness Community Sentiment & Lexical Analysis: Decoding "Gym Talk" on Reddit

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Introduction

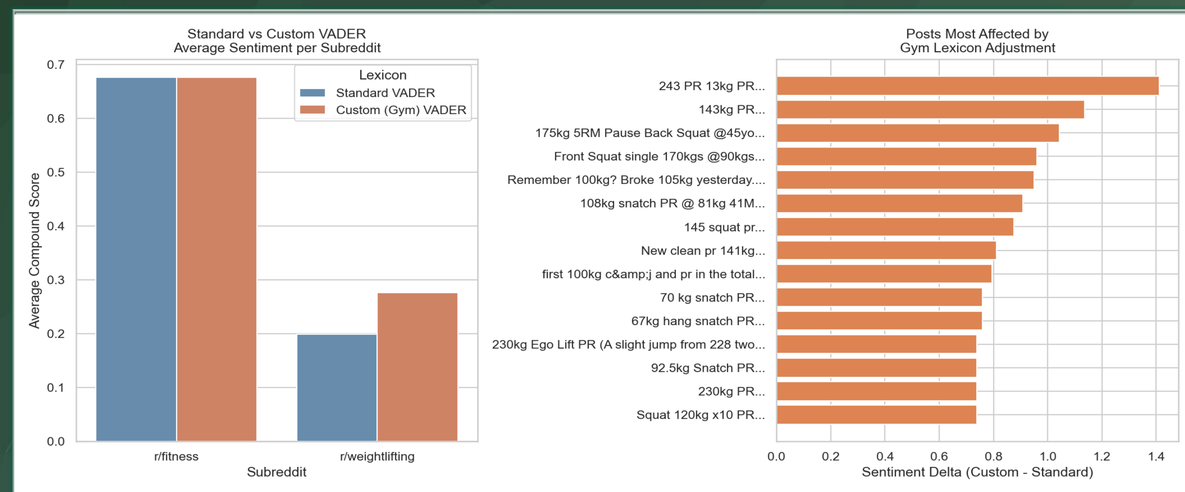
Reddit is one of the largest online communities for fitness discussion. Subreddits like r/fitness and r/weightlifting host thousands of posts monthly, creating a rich dataset for natural language analysis. Analysis into:

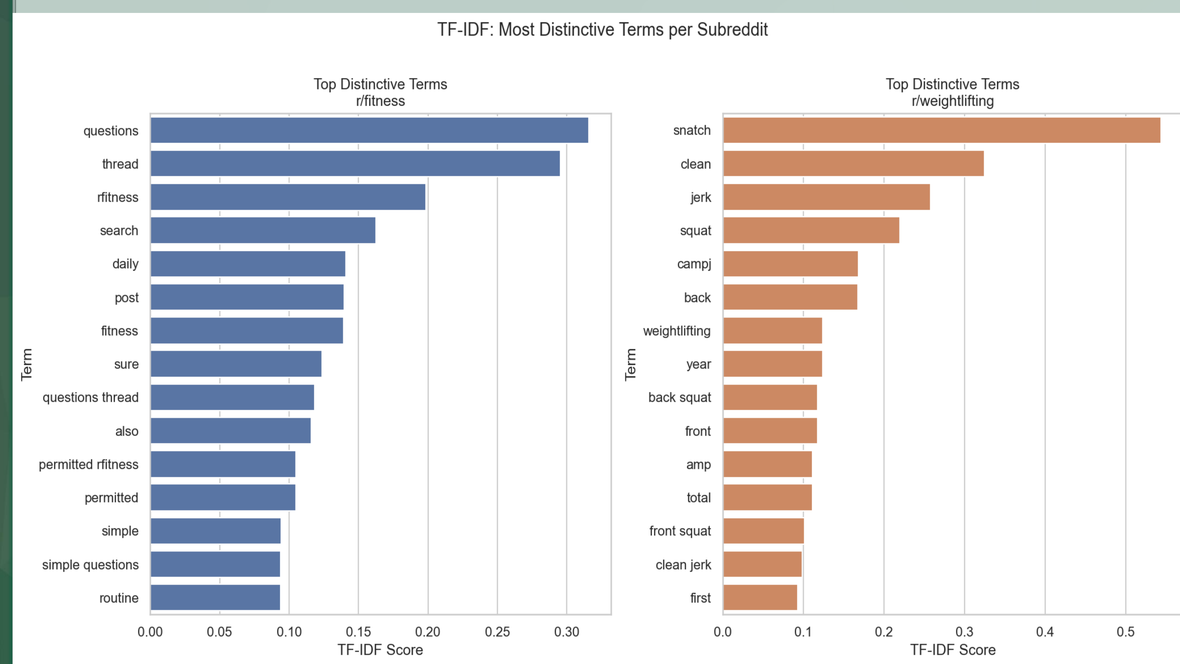
1. Change in Sentiment Analysis when Vader Lexicon was altered to allow for difference in gym lingo.
2. Change in Sentiment analysis over a year.
3. If there was a difference between how information is presented in the subreddits.

Methodology

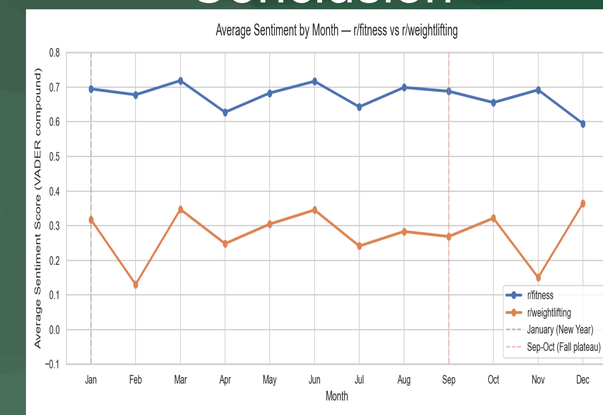
- Data Collection: Reddit public JSON API. Scraped 994 posts and 2,114 comments from r/fitness and r/weightlifting spanning April 2025 – April 2026.
- Text Cleaning: Lowercasing, URL removal (regex), punctuation stripping, tokenization and stopword removal using NLTK.
- Sentiment Analysis: VADER run twice per post: standard lexicon vs. custom gym lexicon
- Lexical Analysis: TF-IDF via scikit-learn treats each subreddit as one document, surfacing the most distinctive terms per community.

Results





Conclusion



This study examined sentiment and lexical patterns across two Reddit fitness communities. Three key conclusions emerge. First, standard sentiment tools like VADER systematically underestimate positivity in specialized communities — r/weightlifting sentiment increased by 39% when domain-specific slang was correctly scored, while r/fitness remained unaffected, suggesting general fitness language aligns more closely with standard English. Second, r/weightlifting demonstrates significantly higher technical density, with top terms drawn from Olympic lifting terminology (snatch, clean & jerk, back squat), while r/fitness discourse centers on community structure and general advice. Third, no seasonal "New Year's Resolution" spike was observed in either community, suggesting that dedicated fitness communities maintain consistent engagement year-round rather than following popular cultural patterns.

Resources

Jupiter Notebook, Python NLP libraries such as NLTK

Acknowledgements

Dr. Charles Hoot
Carson Malen (motivator)

GIT Repo

https://github.com/Sabaroche/WMFinalProject.git